WORKED SOLUTION

STEP III 2016, Question 1

Pure • Reduction Formulae

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q1

The key idea

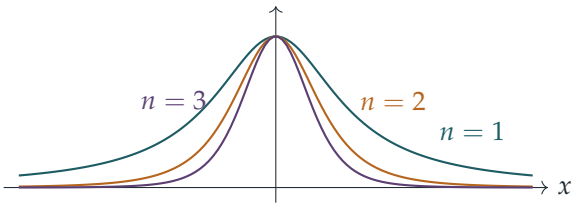
Completing the square turns the quadratic into $(x+a)^2+c^2$, with $c^2 = b-a^2 > 0$, and then the tangent substitution makes I_1 immediate. For the reduction formula, the trick is to integrate by parts *against* 1: differentiating $(x^2+2ax+b)^{-n}$ raises the power to $n+1$, which is exactly what we want to link I_n to I_{n+1}. The induction is then just tidying binomial coefficients.

Solution

Throughout, write

$$Q(x) = x^2+2ax+b = (x+a)^2+c^2, \qquad c = \sqrt{b-a^2} > 0.$$

Since $Q(x) \geq c^2 > 0$ for all x, the integrand is continuous, and since $Q(x)^n$ grows like x^{2n} with $2n \geq 2$, every I_n converges.



The integrands $1/Q^n$ for $a=0$, $b=1$: as n grows they become narrower, so we expect I_n to shrink when $c=1$.

Part (i)

Put $x+a = c\tan u$, so $\mathrm{d}x = c\sec^2 u\,\mathrm{d}u$. As x runs from $-\infty$ to ∞, u runs from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$, and

$$Q(x) = c^2\tan^2 u + c^2 = c^2\sec^2 u.$$

Hence
$$I_1 = \int_{-\pi/2}^{\pi/2} \frac{c\sec^2 u}{c^2\sec^2 u}\,\mathrm{d}u = \frac{1}{c}\int_{-\pi/2}^{\pi/2}\mathrm{d}u = \frac{\pi}{c} = \frac{\pi}{\sqrt{b-a^2}},$$
as required.

Part (ii)

Integrate I_n by parts, taking $u = Q^{-n}$ and $\mathrm{d}v = \mathrm{d}x$. It is cleverer to choose $v = x + a$ rather than $v = x$ (any antiderivative will do), because $Q'(x) = 2(x+a)$. Then
$$\frac{\mathrm{d}}{\mathrm{d}x}\,Q^{-n} = -2n(x+a)\,Q^{-n-1},$$
so
$$I_n = \Big[\,\frac{x+a}{Q^n}\,\Big]_{-\infty}^{\infty} + 2n\int_{-\infty}^{\infty}\frac{(x+a)^2}{Q^{n+1}}\,\mathrm{d}x.$$

? **Ask yourself** Why does the boundary term vanish?

The numerator grows like $|x|$ while the denominator grows like x^{2n}, and $2n \geq 2$ because $n \geq 1$. So $(x+a)/Q^n \to 0$ as $x \to \pm\infty$, and the bracket is zero.

Now use $(x+a)^2 = Q - c^2$ to split the remaining integral:
$$\begin{aligned} I_n &= 2n\int_{-\infty}^{\infty}\frac{Q-c^2}{Q^{n+1}}\,\mathrm{d}x \\ &= 2n\,I_n - 2nc^2\,I_{n+1}. \end{aligned}$$
Rearranging, $2nc^2 I_{n+1} = (2n-1)I_n$, that is,
$$2n(b-a^2)\,I_{n+1} = (2n-1)\,I_n,$$
as required. (Splitting is legitimate because I_n and I_{n+1} both converge.)

Part (iii)

Let $P(n)$ be the statement
$$I_n = \frac{\pi}{2^{2n-2}\,c^{2n-1}}\binom{2n-2}{n-1},$$
where $c^{2n-1} = (b-a^2)^{n-\frac{1}{2}}$.

Base case. For $n = 1$ the right-hand side is $\dfrac{\pi}{2^0 c^1}\binom{0}{0} = \dfrac{\pi}{c}$, which equals I_1 by part (i). So $P(1)$ holds.

Inductive step. Suppose $P(n)$ holds for some $n \geq 1$. By part (ii),
$$I_{n+1} = \frac{2n-1}{2nc^2}\,I_n = \frac{2n-1}{2n}\cdot\frac{\pi}{2^{2n-2}\,c^{2n+1}}\binom{2n-2}{n-1}.$$

We want to show that this equals $\dfrac{\pi}{2^{2n}c^{2n+1}}\binom{2n}{n}$, which is $P(n+1)$. The link between the two binomial coefficients is

$$\binom{2n}{n} = \frac{(2n)!}{n!\,n!} = \frac{(2n)(2n-1)}{n\cdot n}\cdot\frac{(2n-2)!}{(n-1)!\,(n-1)!} = \frac{2(2n-1)}{n}\binom{2n-2}{n-1}.$$

Ask yourself How do we make the powers of 2 match?

Write $\dfrac{2n-1}{2n} = \dfrac{4}{2^2}\cdot\dfrac{2n-1}{2n} = \dfrac{1}{4}\cdot\dfrac{2(2n-1)}{n}$. Then

$$\begin{aligned} I_{n+1} &= \frac{1}{4}\cdot\frac{2(2n-1)}{n}\cdot\frac{\pi}{2^{2n-2}\,c^{2n+1}}\binom{2n-2}{n-1} \\ &= \frac{\pi}{2^{2n}\,c^{2n+1}}\cdot\frac{2(2n-1)}{n}\binom{2n-2}{n-1} = \frac{\pi}{2^{2n}\,c^{2n+1}}\binom{2n}{n}. \end{aligned}$$

Since $2n = 2(n+1)-2$, $n = (n+1)-1$ and $c^{2n+1} = (b-a^2)^{(n+1)-\frac{1}{2}}$, this is exactly $P(n+1)$.

By induction, $P(n)$ holds for every positive integer n:

$$I_n = \frac{\pi}{2^{2n-2}(b-a^2)^{n-\frac{1}{2}}}\binom{2n-2}{n-1}.$$

Checks

- With $a=0$, $b=1$ and $n=2$ the formula gives $I_2 = \frac{\pi}{4}\binom{2}{1} = \frac{\pi}{2} \approx 1.5708$. A numerical evaluation of $\int_{-\infty}^{\infty}(1+x^2)^{-2}\,\mathrm{d}x$ agrees to ten decimal places.
- The tangent substitution also gives $I_n = c^{1-2n}\int_{-\pi/2}^{\pi/2}\cos^{2n-2}u\,\mathrm{d}u$, a standard Wallis integral, whose known value matches the formula.
- For several random a and b with $b > a^2$, numerical integration confirmed the formula for $n = 1,\ldots,5$, and hence also the recurrence of part (ii).

Common slips

- Forgetting to justify that the boundary term in part (ii) vanishes; it needs $n \geq 1$.
- Choosing $v = x$ instead of $v = x+a$ in the integration by parts. This still works, but it leaves an extra $a\int(x+a)Q^{-n-1}\,\mathrm{d}x$ term to deal with.
- Losing track of the power of $b-a^2$ in the inductive step: each step multiplies by $1/c^2$, so the exponent of c goes up by exactly 2, from $2n-1$ to $2n+1$.

Beyond STEP

Integrals over the whole real line of rational functions are a showcase for complex analysis: the residue theorem evaluates I_n in a few lines by looking at the single pole of $1/Q^n$ in the upper half-plane, at $z = -a + ic$. The central binomial coefficients that appear here also turn up in the Wallis product for π and in random walks. Needham's *Visual Complex Analysis* gives a beautiful, geometric account of residues.

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